[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

HOSE AND HOSE ASSEMBLIES, NON-METALLIC (RUBBER, PLASTIC)

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description covers hose and hose assemblies made from either plastic or rubber for use in gardening, nurseries or factory grounds maintenance, general purpose rubber hose, and soaker hoses.

2 CLASSIFICATION.

2.1 Hose and hose assemblies. Hose and hose assemblies will be of the following types, classes, grades, sizes, and styles as specified (see 6.2).

Type I - Rubber, yarn-reinforced hose.

Class 1 - 125 pounds per square inch (psi) working pressure.
Class 2 - 200 psi working pressure.
Size 5/8 - 0.624 inch inside diameter (ID).
Size 3/4 - 0.750 inch ID.

Beneficial comments, recommendations, additions, deletions, and any data which may improve this document should be addressed to: Officer In Charge (Code 15E2), Seabee Logistics Center, 4111 San Pedro St., Port Hueneme, Ca 93043-4410

AMSC N/A FSC 4720

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Type II - Plastic hose.

Grade A - Yarn-reinforced.
Grade B - Non-reinforced.
Size 1/2 - 0.500 inch ID.
Size 5/8 - 0.625 inch ID.
Size 3/4 - 0.750 inch ID.

2.1.1 Soaker hoses. The soaker hoses will be of the following type, and styles, as specified (see 6.2).

Type III

Style 1 - Soaker hose with both male and female couplings.
Style 2 - Soaker hose with a female coupling and an end-closure.

3. SALIENT CHARACTERISTICS.

3.1 First article. When specified (see 6.2), the contractor shall furnish a complete hose length or hose assembly for first article inspection and approval (see 4.2.1 and 6.3).

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.2.1 Rubber. The basic compound shall be natural rubber, synthetic rubber, or a mixture of natural and synthetic rubber. All rubber, tubes and covers shall resist aging.

3.2.2 Plastic. The plastic compound used in the construction of type II, regular hose, and type III, soaker hose, shall be based on resins made by the polymerization of vinyl chloride or copolymerization of vinyl chloride with minor amounts (not over 50 percent) of other unsaturated compounds, suitable plasticized and pigmented.

3.3 Yarns. Reinforcement yarns shall be staple or filament yarns, braided, knitted, or spiral wound.

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3.3.1 Type I. Type I hose shall consist of an all rubber tube and a cover which resists aging and abrasion. Reinforcement shall consist of a braided, knitted or spiral wound cord, for single or double reinforcement.

3.3.2 Type II. Grade A hose shall consist of a plastic tube, a braid or other yarn type belted radial reinforcement and an opaque cover. Grade B hose shall be a plastic tube and cover fused together so that manual separation is not possible.

3.3.3 Type III. Soaker hose shall have a wall thickness of not less than 0.012 inch. The hose shall consist of two or more contiguous chambers of such a profile to insure the hose will lie flat when in use. Soaker openings shall be uniform in size and evenly distributed along the length of each hose. The hose material shall be in accordance with 3.2.1 or 3.2.2.

3.3.4 Lengths. Unless otherwise specified (see 6.2) hose and hose assemblies shall be furnished in standard commercial 25, 50, 75, and 100 foot lengths, excluding couplings. Soaker hoses shall be furnished in 25 and 50 foot lengths.

3.3.5 Diameters. The ID of the hose shall be as specified for the nominal size (see 2.1).

3.4 Fittings (couplings and end cap). Unless otherwise specified (see 6.2), all hose assemblies shall be furnished with commercial type fittings of brass or die casting of zinc base alloys as hereafter specified, attached to the ends. Zinc base alloy die castings shall have a corrosion-resisting coating.

3.4.1 Type I class 1, type II, and type III style 1 hose assemblies. Couplings for type I class 2, type II, and type III style 1 hose assemblies shall be male on one end and female on the other. The male end of type III style 1 hose assemblies shall be furnished with a removal cap. A suitable washer may be inserted in the female coupling or may be secured to the hose.

3.4.2 Type I class 2 hose assemblies. Couplings for type I class 2 hose assemblies shall be male on one end and female on the other. A rubber washer shall be inserted in the female end or may be secured to the hose. Couplings shall be machined or cast brass. Couplings shall be short shank secured with a crimped ferrule, or shall be internal expansion type. The coupling bore shall be full size offering no restriction to the flow of water.

3.4.3 Type III, style 2. Type III style 2 soaker hose shall have a female coupling and a rubber washer on one end. The other end shall have an end-closure fitting accomplished by mechanical means, which shall remain in place, or shall be permanently sealed to form an end closure. The mechanical type end closure shall be removable and replaceable without the aid of special tools.

3.5 Threads. Threads on hose fittings shall be the garden hose type described in FED-STD-H28 except that the thread form and number of threads per coupling shall be in accordance with the manufacturer’s standard commercial product.

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3.6 Physical requirements. Hose shall meet the physical requirements specified in table I. Hose and hose assemblies shall not break or crack when tested in accordance with 4.4.6 at the temperatures specified in table I.

TABLE I. Physical requirements.

	Type I		Type II			
Physical requirements	Class 1	Class 2	Grade A	Grade B	Type III	Test Paragraph
Pressures, pounds per square inch (psi):						
Maximum operating	125	200	-	-	-	- - -
Hydrostatic proof test	150	400	150	100	-	4.4.1
Hydrostatic burst test	500	800	300	200	-	4.4.2
(Adhesion between all components)						
Friction, minimum, pounds	6	6				4.4.3
Tensile strength before aging, minimum pounds per square inch:						
Tube	500	500	-	-	-	4.4.4
Cover	500	500	-	-	-	4.4.4
Tensile strength after Aging percent minimum:						
Tube	60	60	-	-	-	4.4.5
Cover	60	60	-	-	-	4.4.5
Ultimate elongation before aging, percent minimum:						
Tube	150	150	-	-	-	4.4.4
Cover	150	150	-	-	-	4.4.4
Ultimate elongation after aging, percent minimum:						
Tube	60	60	-	-	-	4.4.5
Cover	60	60	-	-	-	4.4.5
Low temperature flexibility	-40° ± 5°F	-40° ± 5°F	32°F	41°F	32°F	4.5.6

3.7 Coupling, requirements. Couplings shall not leak, slip, or blow off of the hose when tested in accordance with 4.4.7.

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3.8 Marking. Hose and hose assemblies shall be permanently and legible marked by commercial methods to identify the manufacturer.

3.9 Workmanship. The hose shall be free of blisters, buckled plies, cracks, laps, and porosity. Couplings shall be free of cracks, dented surfaces and sharp edges. Uncoupled hose in bulk quantities, if specified, shall be free from any defects in workmanship and materials which may limit the use for professional lawn and garden work or the function of heavy duty water hose.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Expect as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements unless disapproved by the Government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection see (4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one sample hose length or hose assembly when a first article is required (see 3.1, 6.2 and 6.3). This inspection shall include the examination of 4.4 and the tests of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the CID and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.4, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with ANSI Z1.4. All hose and hose assemblies of the same type, class, grade, style, length, and size offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete re-inspection. Resubmitted lots shall be re-inspected using tightened inspection. If the rejected lot was screened, re-inspection shall be limited to the defect causing

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rejection. If the lot was reprocessed, re-inspection shall be performed for all defects. Reflected lots shall be separate from new lots, and shall be clearly identified as re-inspected lots.

4.3.1 Sampling for examination. Examination of the hose shall be based on inspection level II and Acceptable Quality Level (AQL) of 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-1 and an AQL of 4.0 percent defective.

4.4 Examination. Each of the sample hose lengths selected in accordance with 4.3.1 shall be visually and dimensionally examined to determine conformance with all the requirements of this specification not involving tests. The classification of defects in FED-STD-162 shall be used to determine and evaluate defects through visual inspection.

4.4.1 Hydrostatic proof pressure test. Test types I and II hose assemblies for visual signs of leakage and damage affecting serviceability at the proof pressures specified in table I. Test procedures shall be in accordance with ASTM D380.

4.4.2 Hydrostatic burst test. Test types I and II hose for hydrostatic strength to withstand burst pressures specified in table I. Test procedures shall be in accordance with ASTM D380.

4.4.3 Friction (adhesion test). Test type I hose for friction requirement specified in table I in accordance with ASTM D380.

4.4.4 Tensile strength and elongation, tube and cover, type I hose only. The tensile strength and elongation of test specimens shall be prepared and tested in accordance with ASTM D412 for both original and aging requirements.

4.4.5 Accelerated aging, type I hose only. Specimens of tube and cover shall be aged in accordance with the method described in ASTM D573 for a period of 70 hours at a temperature of 158° ± 5° Fahrenheit (F) after which they shall be tested in accordance with 4.4.4.

4.4.6 Low-temperature flexibility test (see 3.6).

4.4.6.1 Types I and II hose. Condition a new specimen of hose at least 24 inches long at the temperature specified in table I for 5 hours. Bend the conditioned sample through 180° around a mandrel with an OD of 10 times the ID of the hose within a time period not to exceed 8 seconds. After bending and allowing to return to 70° ± 5°F, the sample shall be hydrostatically tested at 100 psi for a period of 1 minute. Release the pressure and examine the test specimen for conformance to 3.6. Failure to pass this test shall be cause for rejection.

4.4.6.2 Type III hose. A hose specimen approximately 18 inches long shall be exposed to a temperature of 32° ± 2°F for 4 hours in low temperature cabinet or room. At the end of the exposure period, the hose shall be bent through an arc of 180° around 3 ± 1/8 inch diameter

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mandrel at a uniform rate in approximately 5 seconds. The bending and exposure of the specimen shall take place in the same atmosphere. The specimen shall show no cracking or breaks. Failure to pass this test shall be cause for rejection.

4.4.7 Coupling test, types I and II hose assemblies.

a. Test specimen. A representative 12-inch length of hose shall be selected and assembled with male and female couplings conforming to this specification. Couplings shall be attached to the 12-inch sample in accordance with the manufacturer's standard practice.

b. Coupling pull off test. Apply an axial force of 150 pounds (lb) to coupling of the test specimen. The load shall be applied using a suitable tensile test machine or by application of a dead weight. The load shall be held for a period of 1 minute minimum. Slippage of the coupling in excess of 1/16 inch, complete pull out, of rupture of the hoses shall be considered a failure. Following successful completion of the 150 lb loading test, the assembly shall be proof tested at 100 psi for type I class I and type II assemblies. Any leak, further movement of the coupling or blow out shall be considered a failure. The lot represented by the test specimen shall be rejected (see 3.7).

4.5 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.6 Preparation for delivery inspection. An examination shall be made to determine whether the packaging, packing, palletization, and marking comply with the requirements in section 5 of this specification. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an (AQL of 4.0 defects per hundred units).

4.7 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of ASTM SI-10 (IEEE/ASTM SI-10), and all other requirements of this commercial item description including form, fit, and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packing. Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Intended use. The hose and hose assemblies, and soaker hoses covered by the specification are intended for use in the watering and soaking of lawns or gardens and other cleaning tasks that may require water. Type I, class 2 hose and hose assemblies covered by this specification are intended for general purpose use.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, grade, style and size (as applicable) required (see 1.2).
c. If first article is required (see 3.1, 4.2.1 and 6.4).
d. Length required (see 3.3.4).
e. Coupling other than specified (see 3.4).
f. Level of preservation and level of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one hose or hose assembly. The contracting office should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Source of documents.

6.4.1 Government documents are available from the Defense Automated Printing Services, Attn: DoDSSP, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 16111-5094

6.4.2 ASTM documents are available from the American society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19103.

6.4.3 IEEE documents are available from the Institute of Electrical and Electronics Engineers, IEEE Service Center, 445 Hoes Lane, P. O. Box 1331, Piscataway, NJ 08855-1331.

MILITARY INTEREST:

Custodians:
Navy -YD1

Review Activity:
DLA - CC
Navy - CG

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
TVA-TVA
USDA-AFS

Preparing Activity:
Navy - YD1

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